Cooperative Perception with DiscoNet

Course Project Proposal

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MO809 - Topics of Distributed Computing

Objective

Study how vehicles can share information to detect objects that are difficult to see individually. The project asks: **How does DiscoNet compare with no cooperation and early fusion under perfect communication conditions?**

Selected Paper

The distilled collaboration network (DiscoNet) [2] combines information extracted separately by several vehicles. Its collaboration graph assigns importance to each vehicle's information at different locations. During training, knowledge distillation (KD) helps it learn from an early-fusion teacher that sees the combined raw observations. Only the student model is needed for evaluation.

Scope

Use the authors' implementation, with pretrained models where available, on a small, fixed set of held-out scenes from V2X-Sim [1], a vehicle-to-everything (V2X) simulation dataset. Assume **perfect communication settings**: synchronized observations, accurate vehicle poses, and communication without delay or packet loss. Baseline models may be trained when needed to reproduce the reference comparisons. Developing new methods is outside the scope. Vehicles will be simulated as cooperating agents on one computer.

Simple Experiment

Compare DiscoNet with local perception without cooperation and early fusion using compatible checkpoints and identical test scenes. These baseline strategies are also studied in the OPV2V benchmark [3]. Measure detection quality with Average Precision (AP) at Intersection over Union (IoU) thresholds of 0.5 and 0.7, and show example detections. The planned desktop has 16 gigabytes (GB) of random-access memory (RAM) and an RTX 5060 Ti graphics processing unit (GPU) with 16 GB of video RAM (VRAM). Checkpoint availability and software compatibility will be checked first.

Deliverables

A short report, a presentation, and a reproducible Jupyter notebook containing the setup, a results table, and detection plots. The report will identify the dataset version and subset used; results on a subset or a different release will not be presented as an exact reproduction of the paper's benchmark.

Work Plan

Understand the method and validate the setup; run the comparison; prepare the notebook, report, and presentation.

References

[1] Yiming Li, Dekun Ma, Ziyan An, Zixun Wang, Yiqi Zhong, Siheng Chen, and Chen Feng. V2X-Sim: Multi-agent collaborative perception dataset and benchmark for autonomous driving. *IEEE Robotics and Automation Letters*, 7(4):10914–10921, 2022. https://arxiv.org/abs/2202.08449.

[2] Yiming Li, Shunli Ren, Pengxiang Wu, Siheng Chen, Chen Feng, and Wenjun Zhang. Learning distilled collaboration graph for multi-agent perception. In *Advances in Neural Information Processing Systems*, 2021. https://arxiv.org/abs/2111.00643.

[3] Runsheng Xu, Hao Xiang, Xin Xia, Xu Han, Jinlong Li, and Jiaqi Ma. OPV2V: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication. In *IEEE International Conference on Robotics and Automation*, 2022. https://arxiv.org/abs/2109.07644.